Shahjalal University of Science and Technology, Sylhet.

Field Visit Report

Department of Electrical and Electronic Engineering

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1. Introduction

As part of the academic curriculum of EEE 445 (Power System Protection), the students of the Department of Electrical and Electronic Engineering at Shahjalal University of Science and Technology (SUST) undertook an industrial field visit to the Sylhet 225 MW Combined Cycle Power Plant, located at Kumargaon, operated under the Bangladesh Power Development Board (BPDB), Sylhet. The visit was organized under the supervision of the course teacher, Dr. Mohammad Kamruzzaman Khan Prince, Associate Professor, Department of EEE, SUST.

A combined cycle power plant is one of the most thermodynamically efficient configurations for electricity generation. It integrates a gas turbine cycle (Brayton cycle) with a steam turbine cycle (Rankine cycle), where the exhaust heat from the gas turbine is recovered and used to generate steam, which in turn drives a steam turbine to produce additional power. This synergy significantly elevates the overall plant efficiency compared to a simple open-cycle gas turbine plant, making it an ideal subject of study for power engineering students.

This report presents a detailed account of all the systems, equipment, and processes observed during the visit, supported by technical data gathered on-site and supplemented by discussions with plant personnel.

2. Objectives of the Visit

- To gain practical exposure to a real-world combined cycle power plant and its operational environment.
- To understand the integration of gas turbine and steam turbine cycles for improved thermal efficiency.
- To observe the role of protection systems, control panels, and switchgear in power system protection.
- To study the water treatment, cooling, fuel conditioning, and auxiliary systems of the plant.
- To correlate theoretical knowledge from EEE 445 with live industrial systems and equipment.
- To appreciate the challenges of power generation management, load distribution, and fault recovery.

3. Plant Overview

The Sylhet 225 MW Combined Cycle Power Plant is a major electricity generation facility situated in Kumargaon, Sylhet, adjacent to the Surma River. It is operated by the

Bangladesh Power Development Board (BPDB) and serves as one of the primary sources of electrical power for the Sylhet region and beyond. The plant runs on natural gas as its primary fuel and comprises two principal generation units: a gas turbine unit and a steam turbine unit.

The gas turbine, manufactured by Ansaldo Energia of Italy, has an installed capacity of 150 MW and was commissioned on 21st March 2012. It operates at 3000 RPM with a generating voltage of 15.75 kV and a power factor of 0.85. The steam turbine, manufactured by Shanghai Steam Turbine Co. Ltd of China, has a rated capacity of 75 MW and was commissioned on 14th March 2020. Both turbines use ABB's Symphony Plus distributed control system (DCS) for supervisory control and monitoring.

The combined installed capacity of the plant is 225 MW, with a combined cycle efficiency of approximately 50%, significantly higher than the 34–35% efficiency of the gas turbine operating in simple cycle mode alone. On the day of our visit, the plant was operating at approximately 95 MW, as the steam turbine had been taken offline following a protection-related incident that occurred on 12th January 2025.

Figure: View of the Power Plant Premises

4. Control Room

Our visit commenced at the main control room, the nerve centre of the entire plant operation. The control room is equipped with ABB Symphony Plus Distributed Control System (DCS) workstations, which provide real-time monitoring and supervisory control over all major plant parameters, including turbine speed, generated power output, fuel flow, temperature profiles, cooling water flow, and protection relay statuses.

The plant engineers briefed us comprehensively on the operational philosophy of a combined cycle plant. They explained that the gas turbine operates at a thermal efficiency of approximately 35%, while the steam turbine — driven entirely by exhaust heat recovered from the gas turbine exhaust via the HRSG — contributes an additional 18% efficiency, yielding a combined plant efficiency of approximately 53%. The minimum design benchmark for combined cycle efficiency is set at 50%.

We were also informed about an incident on 12th January 2025, where the steam turbine was damaged due to a circuit breaker malfunction. The fault was attributed to an operational error by the Power Grid Company of Bangladesh (PGCB) — the entity responsible for load dispatch and circuit breaker coordination on the grid side — which resulted in an incorrect breaker operation that caused a fault surge reaching the steam turbine generator, necessitating shutdown and repair. At the time of our visit, only the gas turbine was in service, generating approximately 95 MW against the maximum possible output of 220 MW.

The control room also displayed the real-time electrical parameters of the plant: a generated voltage of approximately 115.75 kV at the HV bus, with a line current of approximately 7,300 A.

Figure: Control Room

5. Gas Turbine System

5.1 Air Compressor

The gas turbine cycle begins with the air compressor. The heavy-duty axial-flow air compressor draws in large volumes of ambient air, compresses it to a high pressure ratio, and delivers it to the combustion chamber. We observed the air intake filters, compressor casing, and the inlet guide vane (IGV) control system, which modulates airflow according to load demand. The compressed air is an essential element for sustaining combustion of the natural gas fuel.

Figure: Gas Turbine Air Compressor Unit

5.2 Combustion Chamber and Gas Turbine

Natural gas is injected into the combustion chamber, where it is mixed with the high-pressure compressed air and ignited. The resulting high-temperature, high-pressure combustion gases expand through the turbine blades, rotating the turbine shaft at 3,000

RPM. The turbine shaft is mechanically coupled to the generator, producing electrical energy. The gas turbine model is AE 94.2 by Ansaldo Energia, Italy, with an associated generator model WY21Z-092.

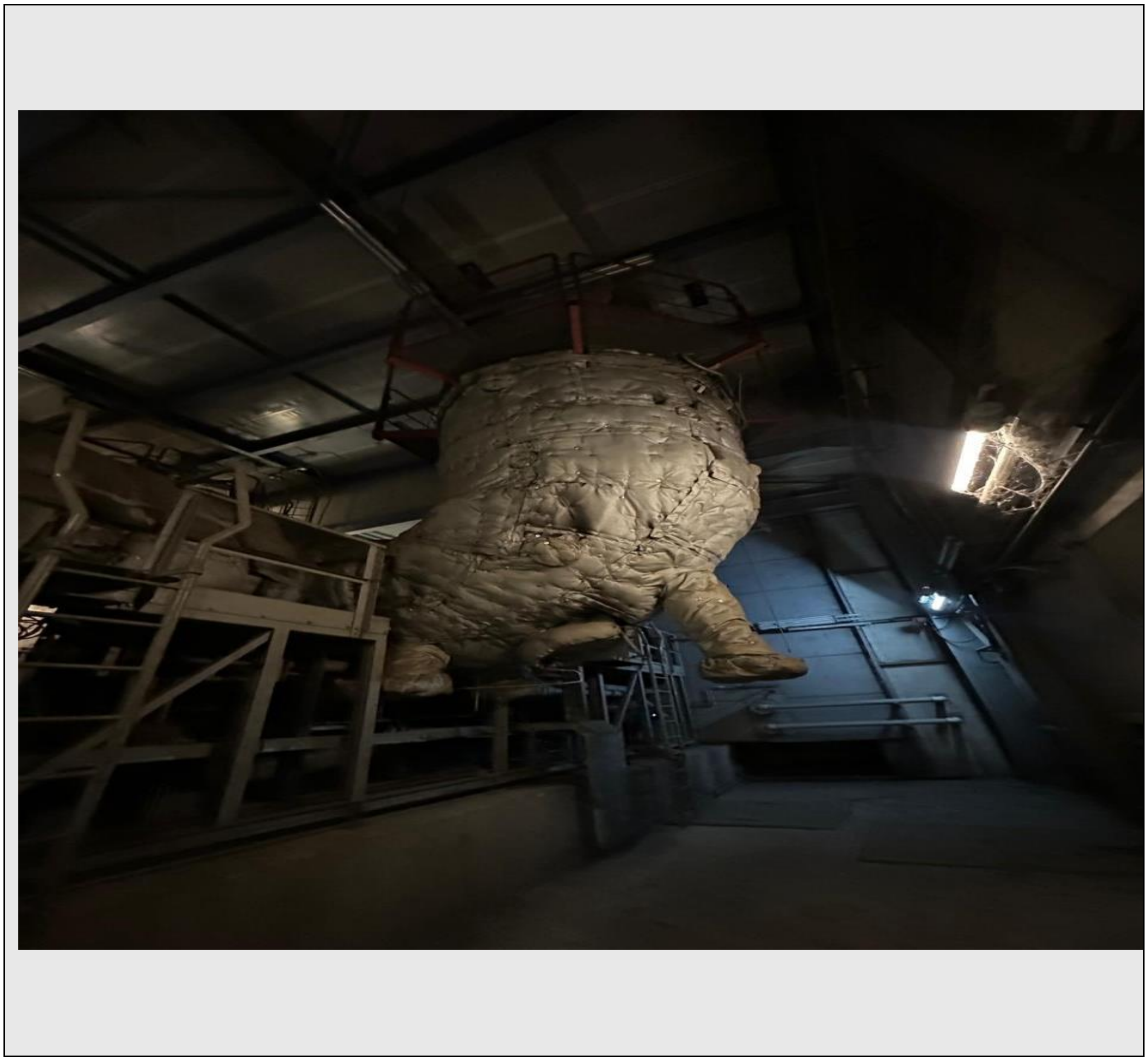

Figure: Gas Turbine

5.3 Diverter Damper

One of the most critical components linking the gas turbine to the HRSG is the diverter damper. This large actuated flap determines whether the hot turbine exhaust gas is directed into the HRSG (for steam generation) or vented to the atmosphere through the bypass stack. During normal combined cycle operation, the damper routes all exhaust gas to the HRSG. When operating in simple cycle mode (as was the case during our visit due to the steam turbine outage), the damper bypasses the HRSG and vents the exhaust gases directly. We observed both the hydraulic, electric, and manual actuation mechanisms of the diverter damper, providing three layers of operational redundancy.

Figure: Diverter Damper with Hydraulic, Electric, and Manual Actuators

5.4 Oil Pump Control System

The gas turbine requires a continuous supply of high-quality lubricating and hydraulic oil to its journal bearings, thrust bearings, and hydraulic systems. We observed the oil pump control panel, which manages the main lube oil pump, the auxiliary oil pump, and the emergency DC-driven oil pump. These redundant systems ensure that oil supply is maintained even during start-up, coastdown, and emergency shutdown conditions, preventing catastrophic bearing failure.

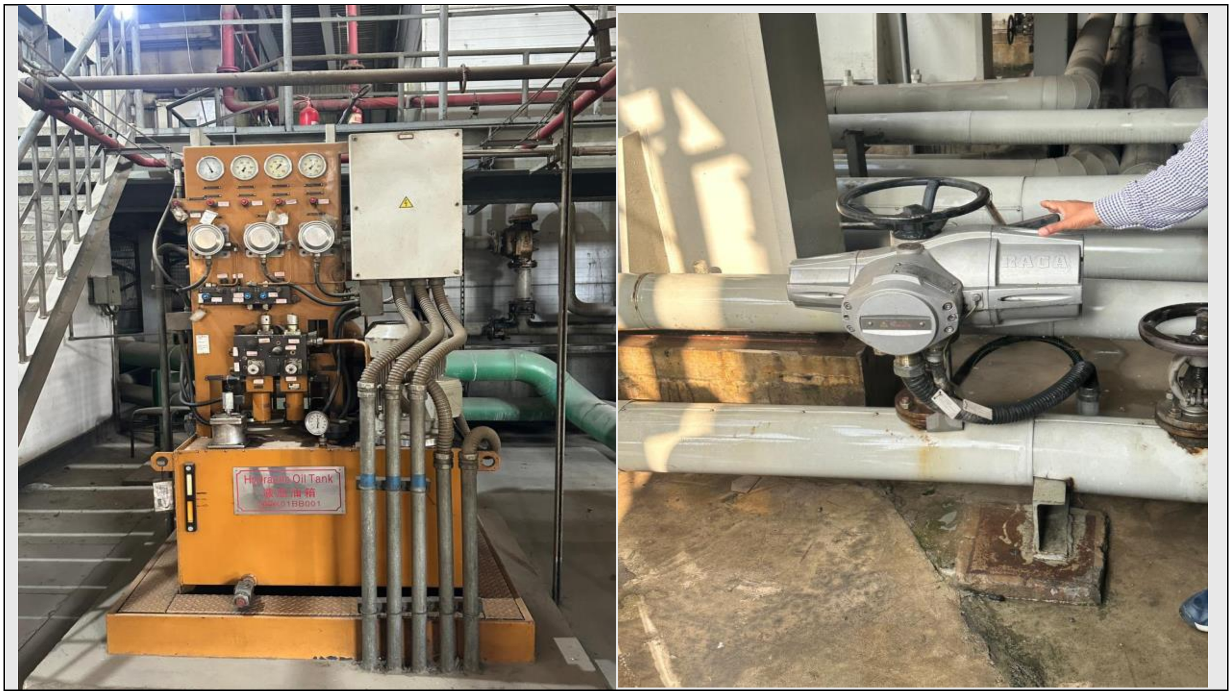

Figure: Oil Pump Control Panel

6. Heat Recovery Steam Generator (HRSG)

The Heat Recovery Steam Generator (HRSG) is the pivotal component that distinguishes a combined cycle plant from a simple cycle plant. Located immediately downstream of the gas turbine exhaust, the HRSG captures the thermal energy from the exhaust gases — which exit the turbine at temperatures in excess of 500°C — and uses it to generate high-pressure steam. The HRSG model at this plant is NG-AE94.2-R, specifically matched to the AE 94.2 gas turbine.

The HRSG we observed is a horizontal, multi-pressure, heat recovery boiler with no supplementary firing. It consists of an economizer section (which preheats the feedwater), an evaporator section (which converts water to steam), and a superheater section (which raises steam to the required temperature and pressure for the steam turbine). The exhaust gas flows over these tube bundles, transferring heat before being discharged through the exhaust stack at a reduced temperature of approximately 80–100°C, thereby minimizing thermal losses.

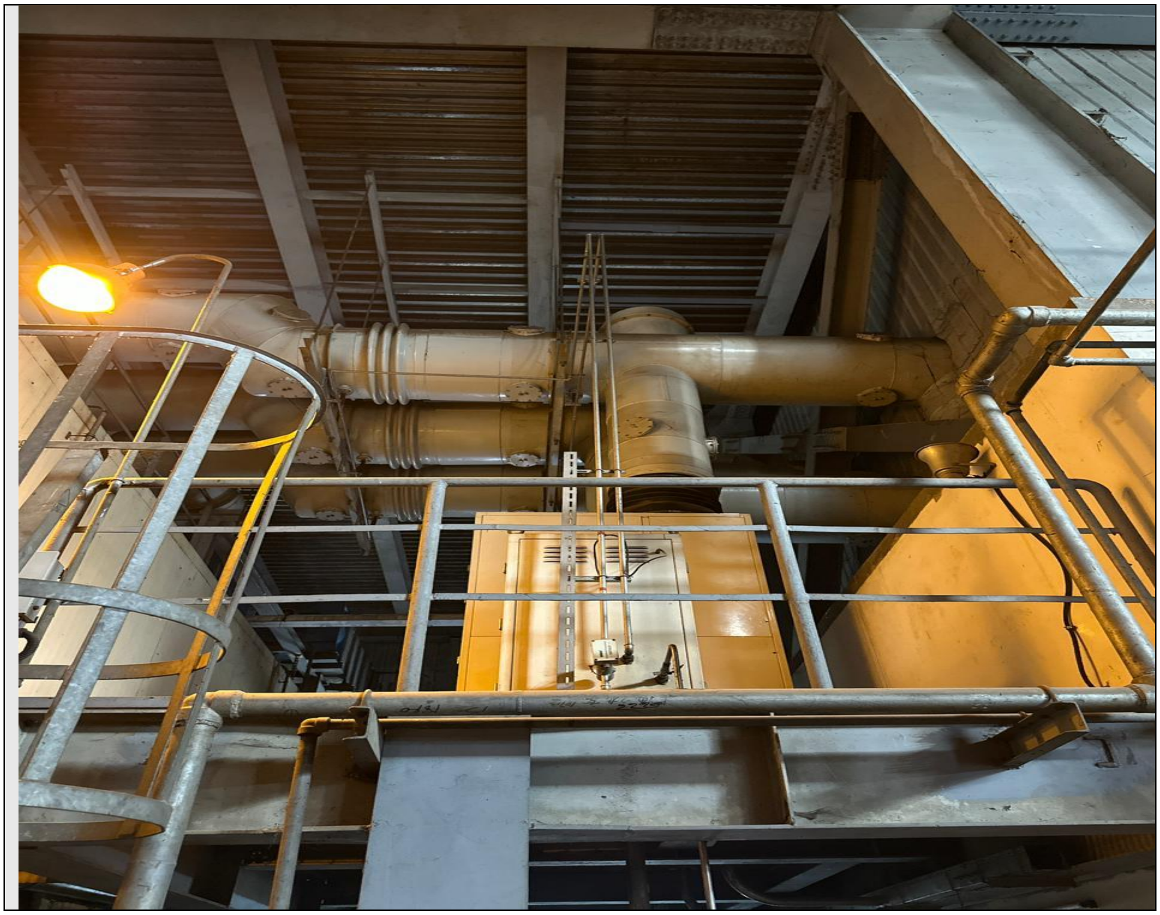

Figure: Heat Recovery Steam Generator (HRSG) – Exterior View

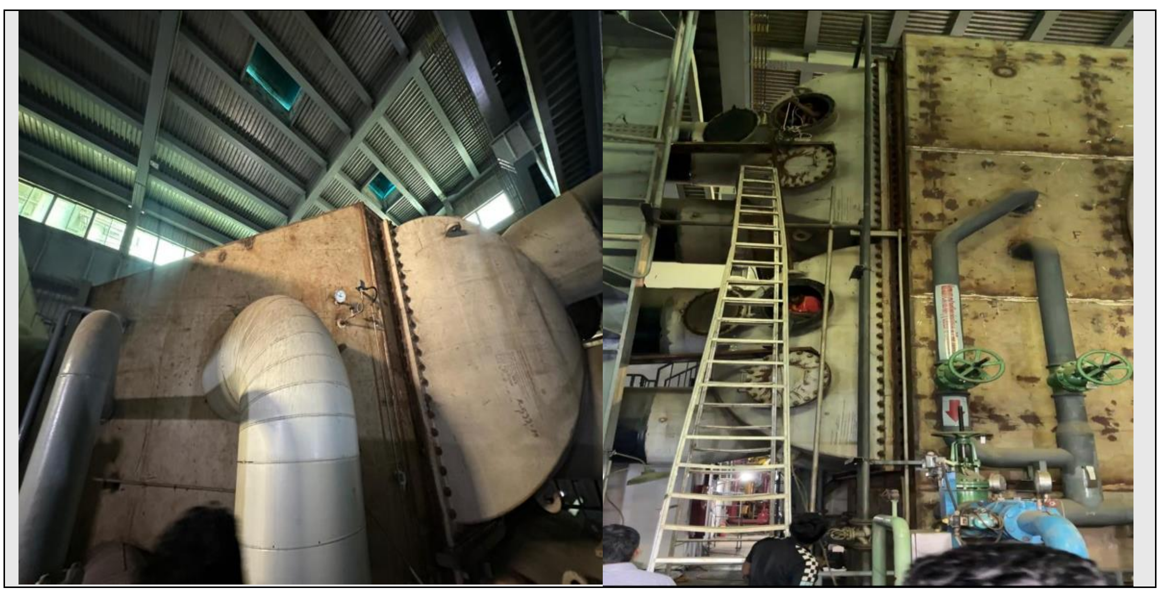

Figure: HRSG Tube Bundles / Drum Assembly

7. Steam Turbine System

7.1 Steam Turbine

The steam turbine, model LNZ88.2 11.12/3.479/[0.431]/542/527.6/[231.4] by Shanghai Steam Turbine Co. Ltd (China), is a condensing steam turbine rated at 75 MW. It operates at 3,000 RPM with a generating voltage of 10.50 kV and a power factor of 0.85. The associated generator model is QF-100-2. The steam turbine was not in operation during our visit due to ongoing repairs following the circuit breaker incident of 12th January 2025, but we were able to observe its physical components and associated systems closely.

Figure: Steam Turbine Assembly (under maintenance)

7.2 Condenser

After the steam expands through the turbine and exhausts its energy, it must be condensed back to liquid water before being fed back into the HRSG as feedwater. The condenser achieves this by passing cooling water (drawn from the Surma River after treatment) through a shell-and-tube heat exchanger. The low-pressure exhaust steam condenses on the outer surface of the cooling tubes and collects in the condenser hotwell as condensate. The condenser maintains a low back-pressure on the steam turbine, maximizing the energy extraction from the steam.

7.3 Condensate Pump Controls

The condensate collected in the condenser hotwell is pumped back to the HRSG feedwater system by condensate extraction pumps (CEPs). We observed the condensate pump control panels, which govern pump start/stop, flow rate, and protection against conditions such as cavitation and low suction head. The condensate then passes through feedwater heaters before being returned as high-pressure feedwater to the HRSG, completing the Rankine cycle.

Figure: Condensate Pump Control Panels

Figure: Condensate Pump Control Panels

8. Cooling Water Treatment Plant

The cooling water system of the power plant is a large and sophisticated infrastructure that ensures a continuous, clean supply of cooling medium for both the condenser and other heat exchangers across the plant. Water is sourced from the Surma River, which flows immediately adjacent to the plant boundary, making this an ideal geographical location for a thermal power plant.

8.1 Water Intake and Treatment

River water is drawn into the plant through intake pumps (CW pumps – Circulating Water pumps) and channelled into the water treatment plant. Here, the raw water undergoes a multi-stage chemical treatment process. Coagulant chemicals (such as alum) are added in coagulant dosing tanks to cause suspended particles and colloidal impurities to clump together. The coagulated flocs then settle out in clarifier tanks through a combination of gravity settling and gentle agitation. We observed these clarifier tanks, which are large open circular or rectangular basins where clarified water overflows at the top while settled sludge is removed from the bottom.

Figure: Water Treatment Plant – Coagulant Dosing Tanks and Clarifier

8.2 Demineralization and Storage

After clarification, the water undergoes demineralization (also referred to as deionization or distillation in this context) to remove dissolved mineral salts that could otherwise cause scaling or corrosion in the HRSG boiler tubes and steam turbine blades. The demineralized water is stored in dedicated storage tanks for use in the boiler/HRSG feedwater circuit.

8.3 Cooling Tower System

The cooling water used in the condenser absorbs heat from the condensing steam and exits at an elevated temperature. This warm water is directed to mechanical draft cooling towers, where large fans draw air upward while water is sprayed downward over fill media. Evaporative cooling reduces the water temperature significantly before it returns to the condenser for another cycle. We observed the cooling tower fans, distribution nozzles, and the collection basin below, where the cooled water falls like rain into the storage basin, giving a visually striking effect of the evaporative cooling process in action.

Figure: Cooling Tower – Fan Assembly and Water Spray System(Inside)

Figure: Cooling water pump, curing chemicals and coagulant tank

9. Fuel (Natural Gas) Treatment System

The plant operates on natural gas supplied through a dedicated pipeline. Before the gas can be admitted into the gas turbine combustion chamber, it must be filtered, conditioned, and compressed to the required pressure. We visited the gas treatment facility at the plant, which comprises a series of gas filter separators that remove liquid droplets, particulate matter, and contaminants from the incoming gas stream.

Gas pressure regulation skids and compressors ensure that gas is delivered at the precise pressure required by the turbine fuel system. At full load, the plant consumes approximately 37 million standard cubic feet per day (mmscfd) of natural gas. The gas treatment system is equipped with safety shut-off valves and pressure relief systems that automatically isolate the gas supply in the event of a fault, an essential safeguard for fire and explosion prevention.

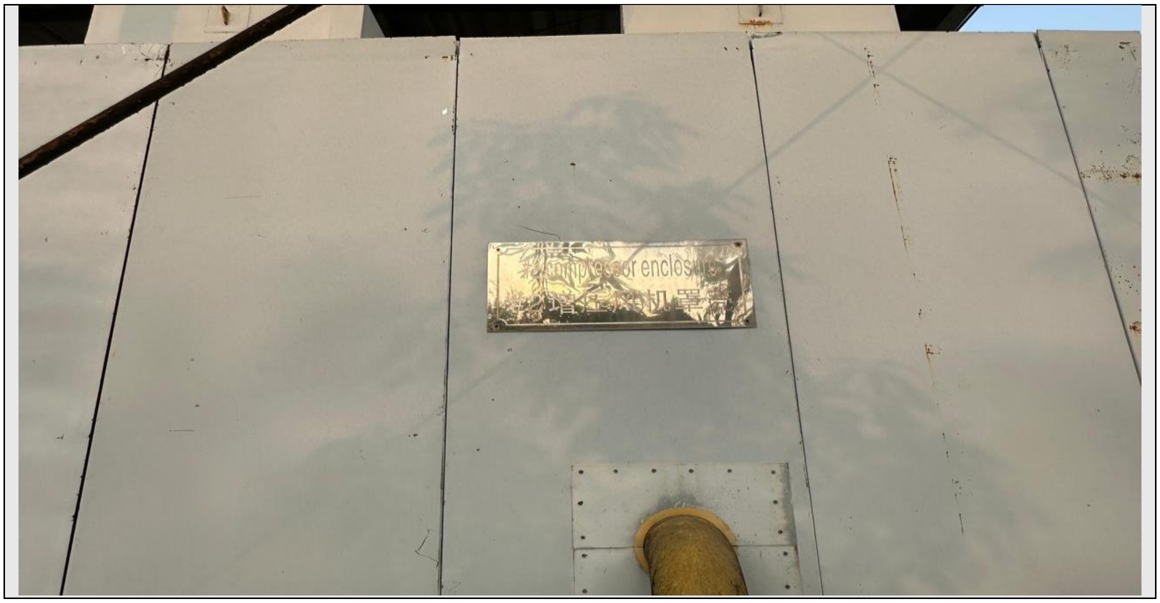

Figure: Natural Gas Filter Separator and Pressure Regulation Skid

10. Electrical Systems and Protection Equipment

10.1 High Voltage (HV) Transformers

The electrical output of both generators (15.75 kV from the gas turbine generator and 10.50 kV from the steam turbine generator) is stepped up to transmission voltage levels by dedicated generator step-up (GSU) transformers. We observed several high-voltage power transformers in the switchyard area. These transformers are oil-cooled and equipped with on-load tap changers (OLTC) for voltage regulation.

Figure: High Voltage (HV) Transformers in the Switchyard

10.2 Transformer Cooling Systems and Oil Level Indicators

Each large power transformer is equipped with an oil cooling system comprising radiator banks, cooling fans, and oil circulation pumps. The transformer oil simultaneously serves as both an insulating medium and a heat transfer medium. We observed the oil level gauges (Buchholz relays and oil temperature indicators), cooling oil checker units, and the radiator assemblies. Buchholz relays are particularly significant from a protection standpoint, as they detect internal faults through gas accumulation and initiate alarms or trip signals.

Figure: Transformer Oil Cooling System and Oil Level Indicators

10.3 Transformer Bushings

We examined the high-voltage bushings of the step-up transformers. Bushings are critical insulating components that allow the high-voltage conductors to pass through the transformer tank wall without the electrical insulation being compromised. The bushings observed were oil impregnated paper (OIP) type for high-voltage applications, with porcelain or composite polymer housings providing environmental protection.


Figure: HV Transformer Bushings

10.4 SFC (Static Frequency Converter) Control Panel

Gas turbines of this class require a Static Frequency Converter (SFC) for startup. The SFC acts as a variable-frequency drive, energizing the generator as a synchronous motor to accelerate the turbine shaft from standstill to firing speed, after which the turbine takes over under its own combustion power. We observed the SFC control panel, which contains power electronic converters (thyristor bridges) and associated control logic.

Figure: SFC (Static Frequency Converter) Control Panel

10.5 Air Circuit Breakers – Intelligent Control Units

We observed the intelligent electronic device (IED) based control units for air circuit breakers (ACBs) used in medium-voltage and low-voltage distribution within the plant. These digital protection and control relays provide over-current, earth fault, and differential protection, with comprehensive event logging and communication capabilities via IEC 61850 or Modbus protocols. The intelligent control units enable remote operation, monitoring, and protection coordination, which is directly related to the subject matter of EEE 445.

Figure: Intelligent Control Units of Air Circuit Breakers

10.6 Sub-Distribution System

The plant houses a comprehensive internal sub-distribution system to supply power to all auxiliary and ancillary loads, including pump motors, fan motors, compressors, lighting, control systems, and HVAC. We visited the main LV switchboard room and observed the distribution board arrangement, motor control centres (MCCs), and bus coupler panels. The bus arrangement provides redundancy so that a fault on one bus section does not de-energize the entire auxiliary supply system.

Figure: Sub-Distribution Panel

11. Auxiliary and Emergency Systems

11.1 Diesel Generator (Emergency Power Supply)

An emergency diesel generator (EDG) is installed at the plant to supply power to critical safety and control loads in the event of a complete loss of station auxiliary power (blackout condition). The EDG automatically starts and picks up critical loads such as emergency lighting, emergency oil pumps, turbine protection systems, UPS systems, and communication equipment. We observed the diesel generator set and its associated automatic mains failure (AMF) panel, which supervises bus voltage and initiates automatic generator startup upon loss of supply.

Figure: Emergency Diesel Generator Set and AMF Panel

12. Key Technical Data Summary

The following table summarizes the principal technical specifications of the plant as gathered during the visit and from the plant information board:

Parameter	Gas Turbine	Steam Turbine
Manufacturer	Ansaldo Energia, Italy	Shanghai Steam Turbine Co. Ltd, China
EPC Contractor	Shanghai Electric Co. Ltd, China	Shanghai Electric Group Co. Ltd (SEC), China
Turbine Model	AE 94.2	LNZ88.2 11.12/3.479/[0.431]/542/527.6/[231.4]
Generator Model	WY21Z-092	QF-100-2
Generating Voltage	15.75 kV	10.50 kV
RPM	3000	3000
Power Factor	0.85	0.85
HRSG Model	—	NG-AE94.2-R
Rated Capacity	150 MW	75 MW
Commissioning Date	21st March 2012	14th March 2020
Control System	Symphony Plus (ABB)	Symphony Plus (ABB)
Fuel	Natural Gas	—
Gas Req. (Full Load)	37 mmscfd	—
Simple Cycle Efficiency	34–35%	—
Combined Efficiency	~50%	—

13. Observations and Discussion

The visit provided an invaluable practical context to the theoretical concepts studied in EEE 445 – Power System Protection. Several key observations merit specific discussion:

Firstly, the importance of protection coordination was brought to life by the steam turbine outage. The incident of 12th January 2025 demonstrated how an incorrect circuit breaker operation by the grid operator (PGCB) propagated a fault to the steam turbine generator, causing equipment damage. This underscores the critical importance of correct relay coordination, zone of protection definition, and inter-utility communication in power system protection.

Secondly, the role of the diverter damper in operational flexibility was well appreciated. The ability to switch between combined cycle and simple cycle operation allows the plant to continue generating power even when the steam side is unavailable, demonstrating robust operational redundancy.

Thirdly, the multi-layered cooling water management system, from river intake through chemical treatment, clarification, demineralization, condenser cooling, and evaporative cooling tower return, illustrated the complexity of auxiliary systems required to support power generation, systems that are often overlooked in academic study but are essential to plant reliability.

Fourthly, the presence of the SFC panel and the emergency diesel generator highlighted the plant's dependence on its own auxiliary power infrastructure — the "black start" capability being a fundamental requirement of any large generating facility.

Finally, the intelligent electronic devices (IEDs) on the air circuit breakers, with their digital communication and event logging capabilities, represented the state of the art in modern power system protection and automation, connecting our coursework on protective relaying directly to real-world implementation.

14. Conclusion

The field visit to the Sylhet 225 MW Combined Cycle Power Plant was an exceptionally enriching educational experience. The plant exemplifies the state of industrial power generation in Bangladesh, combining advanced Italian and Chinese technology under the operational framework of BPDB. The integration of gas and steam cycles, the sophisticated ABB DCS control system, the multi-stage water treatment infrastructure, and the comprehensive electrical protection systems collectively provided a holistic view of what it means to operate a major power plant.

As students of Power System Protection, the most impactful lessons from this visit were the real-world consequences of protection system failures and the elaborate safeguards that are in place to minimize such occurrences. The steam turbine outage, while unfortunate, served as a compelling live case study on the importance of coordinated protection, correct relay settings, and clear demarcation of operational responsibilities between the plant operator and the grid operator.

We express our heartfelt gratitude to Dr. Mohammad Kamruzzaman Khan Prince for organizing this visit, and to the management and engineering staff of the Sylhet 225 MW Combined Cycle Power Plant for their gracious hospitality, detailed explanations, and the invaluable knowledge they shared with us.

15. Group Photo

At the conclusion of the visit, the entire group assembled for a commemorative group photograph within the plant premises. The photograph serves as a record of this academically significant visit and reflects the enthusiasm and engagement of the students and faculty who participated.

Figure: Group Photograph at the Power Plant